Clarity

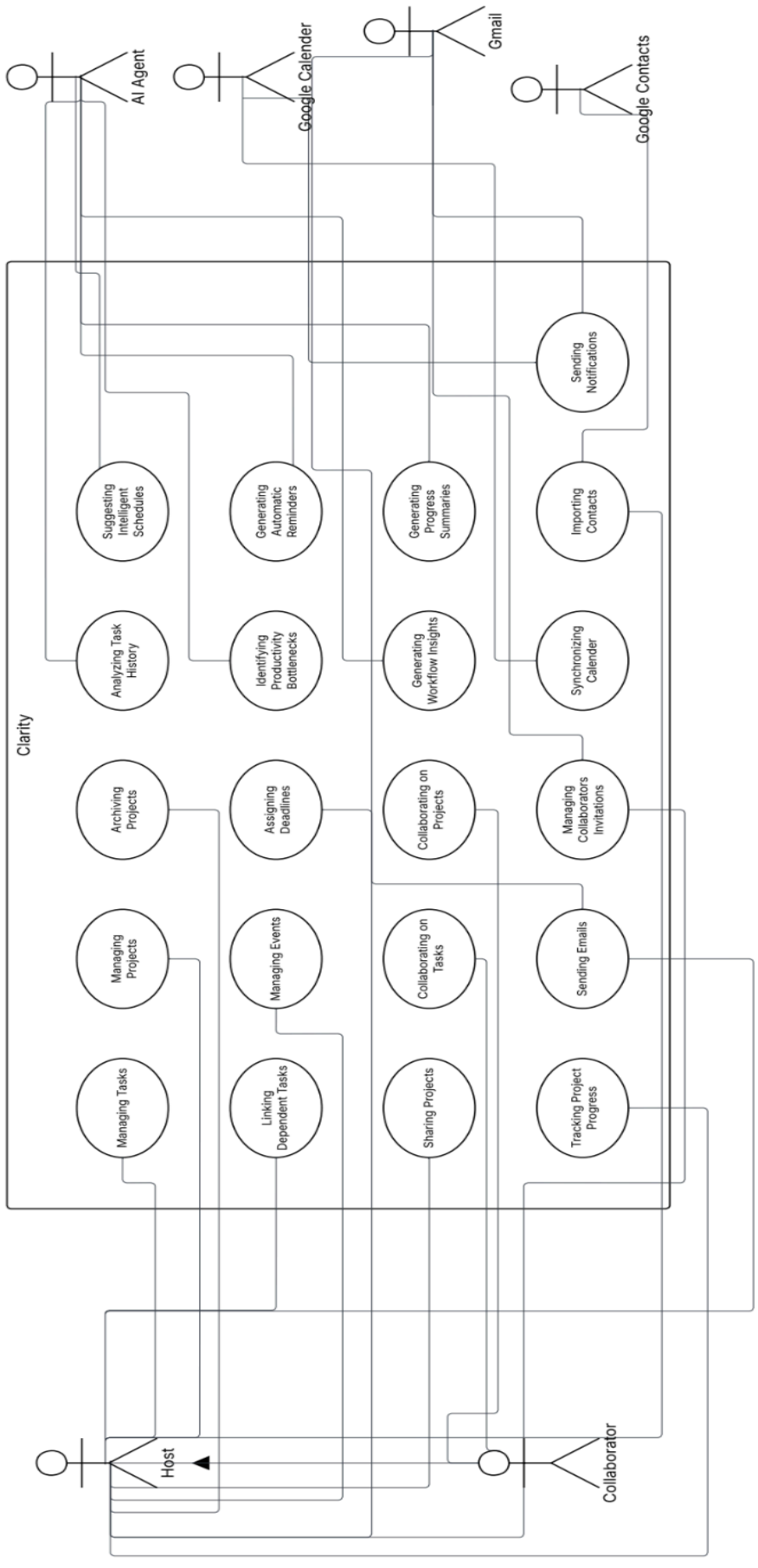
Instructor: Zeeshan Nazar

Section: BCS 5J

Team members

23L-0613 Mawahid Abbas
23L-0757 Bilal Kashif
23L-0848 Mohammad Hamza Iqbal (Team Leader)

Table of contents


AI Agent
Google Calender
Gmail
Google Contacts
Clarity
Suggesting Intelligent Schedules
Generating Automatic Reminders
Generating Progress Summaries
Importing Contacts
Sending Notifications
Analyzing Task History
Identifying Productivity Bottlenecks
Generating Workflow Insights
Synchronizing Calender
Archiving Projects
Assigning Deadlines
Collaborating on Projects
Managing Collaborators Invitations
Managing Projects
Managing Events
Collaborating on Tasks
Sending Emails
Managing Tasks
Linking Dependent Tasks
Sharing Projects
Tracking Project Progress
Host
Collaborator

Identifier	UC-01
Name	Managing Tasks
Summary	This use case describes how the host manages tasks within the system, including creating, editing, deleting, and organizing tasks.
Priority	High
Actors	Host
Pre-condition(s)	Host must be logged into the system. A project or workspace must already exist.
Post-condition(s)	Tasks are successfully updated in the system. All participants can view the updated task list in real time.
Dependencies	Depends on Managing Projects i.e, a project or workspace must exist.

Typical Course of Action

S#	Actor Action	System Response
1	Host selects “Manage Tasks” option.	System displays the task management interface.
2	Host creates a new task by entering details (title, description, deadline).	System verifies and saves the task.
3	Host edits an existing task.	System updates the task details.
4	Host deletes a task.	System removes the task and updates the list.
5	Host organizes tasks by drag-and-drop into categories or priority.	System updates the task order and synchronizes changes in real time.

Alternate Course of Action

S#	Actor Action	System Response
2a	Host enters incomplete or invalid task details.	System prompts host to correct the details.
3a	Host attempts to edit a non-existent or deleted task.	System displays an error message.

4a	Host tries to delete a locked or assigned task.	System displays a warning and prevents deletion.

Exceptions:
System fails to save or update tasks due to server issues.

Identifier	UC-02
Name	Linking Dependent Tasks
Summary	The host links tasks together to establish dependencies (e.g., one task must be completed before another begins).
Priority	Medium
Actors	Host
Pre-condition(s)	Tasks must already exist in the project.
Post-condition(s)	Dependency relationships are created and visible in task flow.
Dependencies	Depends on Managing Tasks as tasks must exist first.

Typical Course of Action

S#	Actor Action	System Response
1	Host selects tasks to be linked.	System highlights selected tasks.
2	Host sets dependency relationship (e.g., Task B depends on Task A).	System records and displays dependency.

Alternate Course of Action

S#	Actor Action	System Response
2a	Host attempts to link a task to itself.	System shows error.
2b	Host tries to link circular dependencies (A→B→A).	System blocks and alerts host.

Exceptions:
Dependency not saved due to server error.

Identifier	UC-03
Name	Sharing Projects
Summary	The host shares a project with collaborators for joint work.
Priority	High
Actors	Host
Pre-condition(s)	Project must exist. Collaborators must have accounts.
Post-condition(s)	Project is shared, and collaborators gain access.
Dependencies	Depends on Managing Projects as a project must exist.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects “Share Project.”	System displays sharing options.
2	Host enters collaborator’s email/ID.	System validates collaborator.
3	Host sets access level (view/edit).	System applies permissions.
4	Host confirms sharing.	System sends invitation to collaborator.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid email ID entered.	System displays error.
3a	Host tries to assign unsupported permission.	System shows warning.

Exceptions:
Email server or notification service unavailable.

Identifier	UC-04
Name	Tracking Project Progress
Summary	The host tracks the project's progress through visual dashboards, analytics, and reports.
Priority	Medium
Actors	Host
Pre-condition(s)	Project and tasks must exist.
Post-condition(s)	Progress data is displayed.
Dependencies	Depends on Managing Tasks and Linking Dependent Tasks as tasks and dependencies provide progress data.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects "View Progress."	System loads dashboard.
2	Host selects metrics (e.g., completion rate, deadlines).	System generates visual reports.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	No tasks exist.	System shows empty progress message.

Exceptions:
Analytics service fails.

Identifier	UC-05
Name	Managing Projects
Summary	The host creates, edits, or deletes entire projects.
Priority	High
Actors	Host
Pre-condition(s)	Host must be logged in.
Post-condition(s)	Project list updated.
Dependencies	No dependency

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects “Manage Projects.”	System displays projects list.
2	Host creates/edits/deletes project.	System updates project records.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Host tries to delete project with active collaborators.	System asks for confirmation.

Exceptions:
Project not saved or deleted due to system error.

Identifier	UC-06
Name	Managing Events
Summary	The host manages events within a project (e.g., deadlines, meetings).
Priority	Medium
Actors	Host
Pre-condition(s)	Project must exist.
Post-condition(s)	Events updated and synchronized.
Dependencies	Depends on Managing Projects as events are tied to projects.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects “Manage Events.”	System shows calendar view.
2	Host creates, edits or deletes event.	System updates calendar.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	User enters an invalid date.	System prompts correction.
2b	User enters an invalid time.	System prompts correction.

Exceptions:
Integration with calendar API fails.

Identifier	UC-07
Name	Collaborating on Tasks
Summary	Collaborators work jointly on tasks, adding comments, making edits, and tracking updates.
Priority	High
Actors	Collaborator
Pre-condition(s)	Project must be shared with collaborator.
Post-condition(s)	Task collaboration history updated.
Dependencies	Depends on Sharing Projects and Managing Tasks as project must be shared and tasks must exist.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Collaborator opens shared task.	System displays task details.
2	Collaborator adds comments or edits task.	System updates and synchronizes changes.
3	Collaborator checks activity log.	System shows updates from all collaborators.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Collaborator tries to edit with read-only access.	System denies and shows warning.

Exceptions:
Synchronization failure due to connection issue.

Identifier	UC-08
Name	Sending Emails
Summary	The host sends emails to collaborators through the integrated Gmail service.
Priority	Medium
Actors	**Primary**: Host **Supporting**: Gmail
Pre-condition(s)	Host must be logged in. Collaborator's email address must be available. Gmail integration must be active.
Post-condition(s)	Email is successfully sent and collaborator receives it.
Dependencies:	Depends on Sharing Projects (project context for communication) and requires Gmail integration.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects "Send Email."	System displays email composition window.
2	Host enters recipient(s), subject, and message.	System validates email details.

3	Host clicks “Send.”	System routes message to Gmail API.
4	Gmail sends the email.	System confirms successful delivery.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid email format entered.	System shows error.
3a	Email exceeds attachment size.	System notifies host.

Exceptions:

Gmail API unavailable.

Internet connectivity lost.

Identifier	UC-09
Name	Archiving Projects
Summary	The host archives completed projects to keep the workspace organized.
Priority	Medium
Actors	Host
Pre-condition(s)	Project must exist.
Post-condition(s)	Project is marked as archived and hidden from active workspace.
Dependencies	Depends on Managing Projects (a project must already exist).
Typical Course of Action	
S#	**Actor Action**	**System Response**
1	Host selects “Archive Project.”	System confirms action.
2	Host approves archive request.	System archives project and moves it to archive folder.
Alternate Course of Action		

S#	Actor Action	System Response
1a	Host cancels request.	Project remains active.

Exceptions:
Archive action fails due to server error.

Identifier	UC-10
Name	Assigning Deadlines
Summary	The host assigns deadlines to tasks or projects.
Priority	High
Actors	Host
Pre-condition(s)	Task or project must exist.
Post-condition(s)	Deadline assigned and reflected in project calendar.
Dependencies	Depends on Managing Tasks (tasks must exist before deadlines can be assigned).

Typical Course of Action

S#	Actor Action	System Response
1	Host selects task/project.	System displays details.
2	Host sets deadline date or time.	System validates entry.

3	Host saves changes.	System updates and displays deadline in calendar.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid date or time entered (past deadline).	System prompts correction.

Exceptions:
Calendar service unavailable.

Identifier	UC-11
Name	Collaborating on Projects
Summary	Collaborators work jointly on shared projects, including editing details, assigning tasks, and tracking updates.
Priority	High
Actors	Collaborator
Pre-condition(s)	Project must be shared with collaborator.
Post-condition(s)	Collaboration changes are reflected in real time.
Dependencies	Depends on Sharing Projects (project must be shared first).

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Collaborator opens shared project.	System displays project details.

2	Collaborator edits or updates project details.	System synchronizes updates.
3	Collaborator views changes from others.	System refreshes project data.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Collaborator has read-only access.	System blocks editing.

Exceptions:
Sync failure due to poor connectivity.

Identifier	UC-12
Name	Managing Collaborator Invitations
Summary	The host invites collaborators to projects via Gmail integration.
Priority	High
Actors	**Primary**: Host **Supporting**: Gmail
Pre-condition(s)	Project must exist. Collaborator must have an email address.
Post-condition(s)	Invitation is sent and collaborator notified.
Dependencies	Depends on Managing Projects (project exists to invite into) and requires Gmail integration.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	Host selects "Invite Collaborator."	System displays invitation form.
2	Host enters collaborator's email.	System validates email.
3	Host sends invitation.	System routes request to Gmail API.
4	Gmail sends invitation email.	System confirms delivery.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Invalid email entered.	System shows error.
3a	Host cancels request.	System does not send invitation.

Exceptions:
Gmail API fails to send invitation.

Identifier	UC-13
Name	Analyzing Task History
Summary	The AI agent analyzes past task data to generate insights.
Priority	Medium
Actors	AI Agent
Pre-condition(s)	Task history must exist.
Post-condition(s)	AI-generated analysis provided to host.
Dependencies	Depends on Managing Tasks as tasks history must exist.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent scans completed task history.	System retrieves stored task data.
2	AI agent applies analysis algorithms.	System generates insights and displays them.
Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Insufficient data provided.	AI informs host that no meaningful analysis can be generated.

Exceptions:
AI service fails.

Identifier	UC-14
Name	Identifying Productivity Bottlenecks
Summary	The AI agent identifies tasks or workflows that cause delays and reduces productivity.
Priority	Medium
Actors	AI Agent
Pre-condition(s)	Task history and progress data must exist.

Post-condition(s)	Host receives suggestions for resolving bottlenecks.
Dependencies	Depends on Analyzing History of Tasks as analysis is prerequisite for identifying bottlenecks.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent reviews task timelines and completion data.	System loads relevant project data.
2	AI agent identifies overdue tasks or frequent blockers.	System highlights bottlenecks.
3	AI agent suggests improvements.	System presents recommendations to host.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	AI finds no bottleneck.	System displays “No issues detected.”

Exceptions:
AI analysis fails due to missing data.

Identifier	UC-15
Name	Generating Workflow Insights
Summary	The AI agent analyzes ongoing tasks and projects to generate insights into workflow efficiency.
Priority	Medium

Actors	AI Agent
Pre-condition(s)	Task and project data must exist.
Post-condition(s)	Workflow insights are displayed to the host.
Dependencies	Depends on Analyzing History of Tasks and Identifying Productivity Bottlenecks.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent scans active projects and tasks.	System retrieves current data.
2	AI agent applies workflow analysis algorithms.	System generates insights.
3	AI agent presents insights.	System displays them in dashboard.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Limited task data.	System shows “insufficient data for insights.”

Exceptions:
AI service error.

Identifier	UC-16
Name	Synchronizing Calendar

Summary	The system synchronizes events and deadlines with Google Calendar.
Priority	High
Actors	Google Calender
Pre-condition(s)	Calendar integration enabled.
Post-condition(s)	Events synced across both systems.
Dependencies	Depends on Managing Events i.e, events must exist to sync.

Typical Course of Action

S#	Actor Action	System Response
1	System prepares event data.	Google Calendar API receives data.
2	Google Calendar confirms sync.	System updates status to host.

Alternate Course of Action

S#	Actor Action	System Response
1a	Duplicate event detected.	System prevents duplicate entry.

Exceptions:
Google Calendar API unavailable.

Identifier	UC-17
Name	Suggesting Intelligent Schedules
Summary	The AI agent suggests optimized schedules for tasks and events based on workload and availability.
Priority	Medium
Actors	AI Agent
Pre-condition(s)	Host must have active tasks/events.
Post-condition(s)	Suggested schedule displayed.
Dependencies	Depends on Synchronizing Calendar and Assigning Deadlines.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent reviews deadlines and availability.	System retrieves relevant data.
2	AI agent generates intelligent schedule.	System displays recommendations.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Conflicting deadlines detected.	System highlights conflicts for manual resolution.

Exceptions:

AI scheduling module fails.

Identifier	UC-18
Name	Generating Automatic Reminders
Summary	The AI agent generates reminders for tasks, events, or deadlines.
Priority	High
Actors	AI Agent
Pre-condition(s)	Tasks or events must exist with deadlines.
Post-condition(s)	Reminder notifications sent to host.
Dependencies	Depends on Suggesting Intelligent Schedules and Managing Events.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent scans upcoming deadlines.	System identifies tasks/events.
2	AI agent generates reminder.	System schedules reminder notification.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	No upcoming deadlines.	System displays “no reminders.”

Exceptions:
Notification service fails.

Identifier	UC-19
Name	Generating Progress Summaries
Summary	The AI agent generates summaries of project progress for host review.
Priority	Medium
Actors	AI Agent
Pre-condition(s)	Project or task data must exist.
Post-condition(s)	Progress summary displayed in dashboard.
Dependencies	Depends on Tracking Project Progress and Analyzing History of Tasks.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	AI agent reviews task completion data.	System aggregates data.
2	AI agent generates summary.	System displays progress summary.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	No completed tasks.	System shows “progress summary unavailable.”

Exceptions:
AI service failure.

Identifier	UC-20
Name	Importing Contacts
Summary	The system imports user contacts from Google Contacts for collaboration and communication.
Priority	Medium
Actors	Google Contacts
Pre-condition(s)	Google Contacts integration enabled.
Post-condition(s)	Contacts successfully imported.
Dependencies	No direct dependency except system login; may indirectly support Managing Collaborator Invitations.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	System requests contacts.	Google Contacts API provides list.
2	System imports contacts.	System updates user's contact directory.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
2a	Duplicate contact found.	System merges or skips.

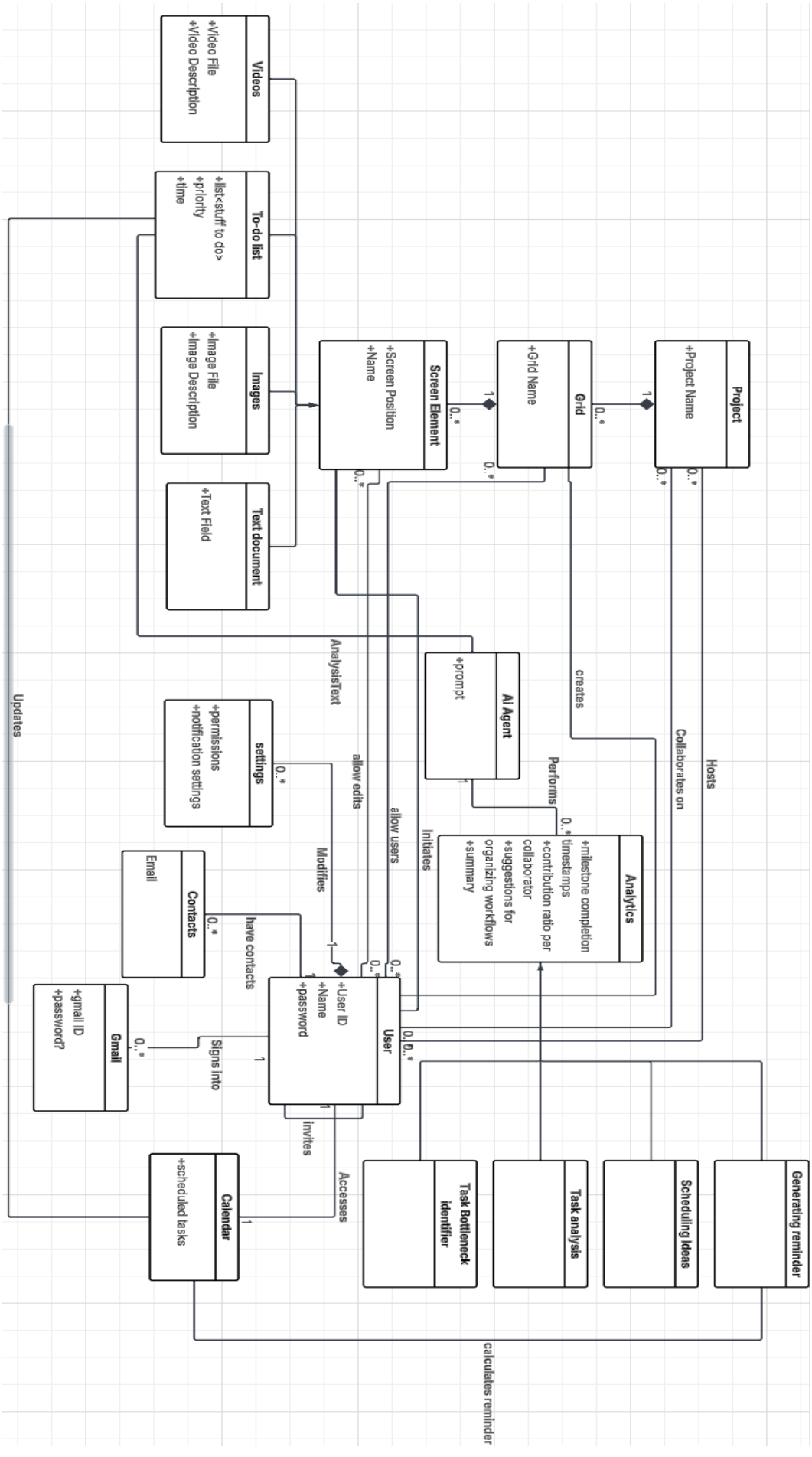
Exceptions:
Google Contacts API unavailable.

Identifier	UC-21
Name	Sending Notifications
Summary	The system sends notifications (e.g., reminders, updates) to users using Gmail and Google Calendar.
Priority	High
Actors	Google Calendar Gmail
Pre-condition(s)	Notifications must be scheduled.
Post-condition(s)	User receives notification successfully.
Dependencies	Depends on Managing Events, Sharing Projects, and external services Google Calendar and Gmail.

Typical Course of Action		
S#	**Actor Action**	**System Response**
1	System generates notification event.	Google Calendar/Gmail API processes request.
2	Notification sent to user.	System confirms delivery status.

Alternate Course of Action		
S#	**Actor Action**	**System Response**
1a	Invalid recipient.	System shows error.

Exceptions:
Gmail API failure.
Google Calendar API failure.


Project
+Project Name
Grid
+Grid Name
Screen Element
+Screen Position
+Name
Videos
+Video File
+Video Description
To-do list
+list<stuff to do>
+priority
+time
Images
+Image File
+Image Description
Text document
+Text Field
AI Agent
+prompt
Analytics
+milestone completion timestamps
+contribution ratio per collaborator
+suggestions for organizing workflows
+summary
settings
+permissions
+notification settings
Contacts
Email
User
+User ID
+Name
+password
Gmail
+gmail ID
+password?
Calendar
+scheduled tasks
Generating reminder
Scheduling Ideas
Task analysis
Task Bottleneck identifier
Hosts
Collaborates on
creates
Performs
Initiates
allow users
allow edits
Modifies
have contacts
Signs into
invites
Accesses
AnalysisText
Updates
calculates reminder